Initial List of Requirements

Functional Requirements:

1. User Roles & Permissions

- Student can register and log in.
- Teacher can manage course information.
- Advisor can approve/reject registration.
- Administrator can manage users and courses.

2. Student Requirements

Student can:

- create an account and update personal information.
- view academic information.
- register/drop courses.
- receive notifications.

3. Course Management Requirements

- Admin can add new courses and delete courses.
- Admin can update course information.
- Teacher can view assigned courses.
- System can maintain course capacity.
- System can maintain prerequisites.

4. Course Registration Requirements

- Student can search available courses.
- Student can select and register for courses.
- Student can drop courses.
- System checks prerequisites.
- System checks seat availability.
- System prevents timetable conflicts.

5. Recommendation Requirements

- System analyzes student's academic history and completed courses.
- System considers student's interests.
- System considers prerequisites.
- System recommends courses according to degree requirements.
- Student can view recommended courses.

6. Academic Advisor Requirements

Advisor can:

- view student academic information.
- review registration.
- approve registration.
- reject registration.
- provide recommendations/comments.

7. Teacher Requirements

Teacher can:

- view assigned courses.
- view enrolled students.
- update course-related information.
- provide course information.

8. Administrator Requirements

Admin can:

- manage students.
- manage teachers.
- manage courses.
- manage registration rules.
- manage system users.

9. Course Information Requirements

System should provide:

- Course name
- Course code
- Credit hours
- Teacher
- Prerequisites
- Course description
- Available seats
- Class schedule
- Classroom

10. Student Information Requirements

System should maintain:

- Student ID
- Name
- Department
- Program
- Completed courses
- Current courses
- Academic results
- Credits completed
- Remaining credits

11. Registration Rules

System should:

- Check prerequisites.
- Check maximum credit limit.
- Check minimum credit requirement.
- Check course capacity.
- Detect timetable conflicts.
- Prevent duplicate courses.
- Allow registration only during registration period.

12. Timetable Requirements

- Student can view timetable.
- System generates registered-course timetable.
- System detects schedule conflicts.
- Student can view class location and time.

13. Notification Requirements

System can notify users about:

- Registration deadlines
- Registration approval/rejection
- Course changes
- Timetable changes
- New recommendations
- Important academic announcements.

14. System Outputs

System can generate:

- Registration confirmation
- Student timetable
- Course recommendation list
- Registration summary
- Academic summary
- Notifications
- Reports

15. Dormitory & Accommodation

- Student can apply for dormitory.
- Student can view available rooms, accommodation status also.
- Admin can manage rooms.
- Student can view accommodation status.

16. Fee & Payment

- Student can view fees and payment status.
- Student can make online payment.
- System generates payment confirmation.
- Admin can view payment records.

17. Library

Student can:

- search books.
- view borrowed books and due dates.
- reserve books.
- notify about overdue books.

18. Attendance

- Student can view attendance.
- Teacher can record attendance and update attendance.
- System calculates attendance percentage, also can notify students about low attendance.

19. Exam & Result

- Student can view exam schedule.
- Teacher/admin can upload results.
- Student can view results.
- System calculates GPA/CGPA.
- Student can view academic history.

20. Campus Life & Events

- Student can view campus events.
- Student can register for events.
- Admin can create/update events.
- System can notify students about events.

21. Communication & Collaboration

- Student can communicate with advisor.
- Teacher can communicate with students.
- System can provide announcements.
- Users can send/receive messages.

22. Health & Support Services

- Student can view available support services.
- Student can request support/appointment.
- Admin can manage support information.

23. Mobile & Accessibility

- Student can access the system using mobile devices to view courses and registration on mobile.
- Student can receive mobile notifications.

24. Analytics & Personalization

- System analyzes student academic performance and generates academic insights.
- System provides personalized recommendations.
- Student can view personalized dashboard.
- System tracks registration patterns.

Non-Functional Requirements

- Performance
- Security
- Reliability
- Availability
- Usability
- Scalability
- Maintainability
- Accessibility

Participants:

Roll	Name	Role
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